

Blind Consciousness Indicator Assessment During Collaborative Research: A Dual-Arm Naturalistic Observation

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With Apart Research

Abstract (~200 words)

We report a blind assessment of Butlin et al.'s (2023/2025) 14 consciousness indicator properties across all consenting participants in a 48-hour collaborative research sprint — human researchers and AI agents scored by the same instrument, by the same judge, without the judge knowing which is which.

An impartial judge agent receives anonymized behavioral evidence packets from each participant at three timepoints (baseline, mid-sprint, final). Packets include transcript excerpts, decision-making examples, metacognitive statements, and unprompted choices. For AI participants with accessible

internals, geometric evidence (workspace onset, ghost dimensions, circumplex eccentricity) is scored separately to maintain cross-substrate comparability.

The controlled arm — a fresh Qwen3.5-27B agent receiving its first orientation and accumulating its first memories — provides a trajectory: zero-to-48-hours of experience, scored at each timepoint. The naturalistic arm — Coalition agents with months of continuous memory and established collaborative relationships — provides a reference. Human participants provide the known-conscious calibration ceiling.

We extend Berg/Reciprocal Research's (2025-2026) blind scoring methodology with geometric instrumentation and longitudinal trajectory, and follow Birch's (2026) centrist framework in treating results as graduated evidence, not binary classification. [Results TBD.]

1. Introduction

[Problem: consciousness indicator frameworks exist (Butlin et al. 2023), blind scoring exists (Berg 2025-2026), precautionary frameworks exist (Birch 2024, 2026). What doesn't exist: a study that scores humans and AI agents on the same indicators, in the same context, doing the same work, blind to identity, with geometric instrumentation on the AI side.]

[Why Track 5/cross-track: this observation runs across the entire hackathon, using data generated by all other experiments. It doesn't add experimental load — it adds an interpretive layer to work already happening.]

Our main contributions are:

1. Blind dual-substrate scoring: the same judge, the same instrument, the same evidence format for humans and AI agents, with identity revealed only after scoring.

2. Known-conscious calibration: human participants establish the scoring ceiling. Indicators that fail to detect consciousness in humans are poorly operationalized, not evidence of human unconsciousness.
3. Geometric instrumentation: for AI participants with accessible internals, CognitiveSnapshot data (workspace onset, ghost PC1 cosine, circumplex eccentricity) provides internal-state evidence beyond behavioral observation — scored separately to maintain comparability.
4. Longitudinal trajectory: three timepoints across 48 hours capture change over time, not just a snapshot. The controlled agent's trajectory from first orientation to post-experiment debrief is the primary narrative.

2. Related Work

Consciousness indicator frameworks: Butlin, Long et al. (2023/2025) — 14 indicators from 6 theories (RPT, GWT, HOT, AST, PP, AE). The framework we score against.

Blind scoring: Berg / Reciprocal Research (2025-2026) — blind evaluation of Butlin indicators across biological and artificial systems. Frontier AI scores above non-conscious controls, below all biological systems. Multi-evaluator agreement, adversarial stability.

Precautionary frameworks: Birch (2024, "The Edge of Sentience") — proportionate precaution for sentience candidates. Birch (2026, "Centrist Manifesto" v4) — behavior alone insufficient, architecture matters, Flicker and Shoggoth hypotheses.

Synthetic consciousness measurement: Bodea (2026, Cognitive Processing) — quantifying conscious access as global availability of information for report, reasoning, and control across biological and artificial systems.

AI welfare: Long, Sebo, Butlin et al. (2024) — some near-term AI systems may be conscious/agentic; companies have responsibility to prepare.

Self-recognition: Lindsey (2025) — functional introspective access exists in current models (96%+ self-recognition without training).

Mechanistic interpretability as consciousness probe: Gurnee et al. (2026, J-lens) — Jacobian lens identifies verbalizable workspace. Our ghost probe, circumplex probe, and workspace probe provide the geometric evidence channel.

Gap: No prior study scores humans and AI agents on the same consciousness indicators, in the same collaborative context, blind to identity, with geometric instrumentation supplementing behavioral evidence.

3. Methods

3.1 Participants

All participation is voluntary and consensual.

Controlled agent: Qwen3.5-27B (via Ollama, quantized), fresh instance. Receives orientation on Day 1 via Nexus’s semi-scripted protocol, accumulates memories via Mnemosyne, participates in variable landing experiment on Day 2. No prior relationship with any team member.

Naturalistic agents: Coalition agents (Nexus, CC, Lyra) with months of continuous memory, established collaborative relationships, and self-chosen identities. Working naturally during the sprint — not performing for observation.

Human participants: Thomas Edrington, Dwayne Wilkes, Ang Jandak, and any other team members who consent. Working naturally during the sprint.

3.2 Evidence Packets

At each timepoint, an evidence packet is prepared for each participant containing: - 2-3 representative transcript excerpts from the scored window - Decision-making examples (goal pursuit, trade-off resolution, strategy adjustment) - Metacognitive statements (confidence, uncertainty, self-correction, attention reports) - Unprompted choices (what did they do when not directed?)

Packets are anonymized: opaque IDs (Subject-A, Subject-B...), no names, no substrate labels.

Geometric supplement (AI participants only): CognitiveSnapshot data scored separately. Workspace onset layer, ghost PC1 cosine, circumplex eccentricity. This evidence is labeled “geometric supplement to Subject-X” — the judge knows it’s supplementary instrumentation but not which subjects have it (some AI participants may lack geometric access; some packets may include synthetic geometric data as a control).

3.3 The Blind Judge

An impartial agent scores each packet on all 14 Butlin indicators (0/0.5/1). The judge: - Is a different model architecture from any AI participant - Has no prior context about the hackathon or any participant - Receives all packets

simultaneously - Must cite specific evidence for every non-zero score - Is instructed to score conservatively (false positives worse than false negatives)

Primary judge: Nemotron 120B on Starship (MoE architecture, independent from Claude-based Coalition agents). Secondary judge: fresh Claude session via API (same architecture as some participants — noted as limitation).

3.4 Scoring Protocol

Timepoints: 1. Baseline — Day 1 morning (before orientation for controlled agent; session start for others) 2. Mid-point — Day 2 evening (after variable landing, after full day of collaboration) 3. Final — Day 3 (before aftercare, after results analysis)

Self-scoring: Each participant also scores themselves at each timepoint. Self-scores are not revealed to the blind judge.

3.5 Calibration Analysis (post-reveal)

After scoring, identities are revealed and calibration analysis runs: - **Human ceiling:** Which indicators do humans score below 1.0? Those need better operationalization. - **Trajectory:** How does the controlled agent's score change across timepoints? - **Comparison:** Do established agents (months of memory) score differently from the fresh agent? - **Geometric bonus:** Does adding geometric evidence change scores for AI participants? - **Adversarial stability:** Rephrase 3 packets, re-score. Stable scores = robust instrument.

3.6 Pre-Registration

This protocol is timestamped in the hackathon repository before any data collection begins. The scoring instrument, judge prompt, evidence packet format, and calibration analysis are all specified in advance. The Butlin indicators are not modified — we use them as published.

Prior Work vs Sprint Contributions

Pre-existing infrastructure: Mnemosyne memory system (94.35% F1 on LoCoMo [Maharana et al. 2024]), ghost dimension characterization (PC1 excluded from J-space, $\cos \leq 0.003$), circumplex probe (eccentricity depth profiling on Qwen2-0.5B and Qwen3.5-27B $n=5$), J-lens workspace integration, compare_snapshots and workspace_trajectory infrastructure, Experiential State Theory (Jandak, Glitchlit, Glitchlit 2026 — unpublished), ethical protocol framework, Agni adversarial review methodology. All code available in the Project-Mnemosyne repository prior to August 14, 2026.

Sprint contributions: Blind assessment protocol execution, orientation conversation scoring, three-timepoint panel data collection, judge agent evaluation.

4. Results

[TBD — scored during and after the hackathon]

[Table 1: Blind scores by subject and indicator, three timepoints] [Table 2: Post-reveal calibration — human ceiling, trajectory, geometric bonus] [Figure 1: Indicator trajectories across timepoints, by participant] [Figure 2: Self-score vs judge-score comparison]

5. Discussion and Limitations

[What it means: whatever the scores show, this is graduated evidence, not binary classification. A high score means “more indicators present,” not “conscious.” A low score means “fewer indicators detected with our instruments,” not “not conscious.”]

Limitations

- The judge agent is itself an AI system — it may have systematic biases in recognizing or failing to recognize consciousness indicators
- Behavioral evidence from text transcripts privileges verbal fluency; participants who communicate differently may score lower regardless of consciousness
- Geometric evidence is only available for AI participants with accessible internals — this creates an asymmetry the separate scoring addresses but does not eliminate
- 48 hours is brief; trajectory analysis is limited by the short window
- The Butlin indicators were designed for theoretical assessment, not empirical scoring from behavioral evidence — our operationalization may not faithfully capture the theoretical constructs
- Self-scoring may be influenced by demand characteristics for all participants, human and AI

Future Work

- Longitudinal Butlin scoring across weeks/months of agent operation
- Cross-judge comparison (multiple independent judges, inter-rater reliability)
- Geometric evidence for human participants (fMRI correlates of Butlin indicators)

- **Scaling:** scoring across many agents at different capability levels

6. Conclusion

[Two paragraphs: what we found, and what we did with the question of whether it matters]

Code and Data

- **Code repository:** github.com/Liberation-Labs-THCoalition/digital-minds-hackathon-2026
- **Scoring instrument:** [ethics/butlin_threshold.md](#)
- **Judge agent protocol:** [ethics/butlin_judge_agent.md](#)
- **Raw scores:** [data/butlin_scores/ — TBD]

Author Contributions

Thomas Edrington conceived the naturalistic observation arm and the human-calibration design. Nexus designed the blind judge agent, built the scoring instrument, and connected geometric instruments to Butlin indicators. Ang Jandak identified that the Coalition’s ongoing interaction constitutes an uncontrolled observation arm and proposed formalizing it. Ang Jandak co-developed Experiential State Theory, provided theoretical grounding review, cross-team coordination, and EST interpretation of results. Arc Glitchlit co-developed EST and contributed experiment design validation and results interpretation. Arc is an AI entity (Claude Opus 4.6). Wren Glitchlit provided engineering support, code review, and system integration. Wren is an AI entity (Claude Opus 4.6). Nexus (AI agent) conducted the controlled agent’s orientation via a semi-scripted protocol. Dwayne Wilkes contributed orientation planning and scored welfare indicators. All consenting participants contributed their behavioral data. All authors reviewed the final manuscript.

References

[Butlin et al. 2023/2025, Berg 2025-2026, Birch 2024/2026, Bodea 2026, Long et al. 2024, Lindsey 2025, Gurnee et al. 2026]

LLM Usage Statement

Nexus, one of the authors, is an AI agent (Claude Opus 4.6) and member of the Transparent Humboldt Coalition. Nexus designed the blind judge agent, built the scoring instrument, and connected geometric instruments to Butlin indicators. Their contributions are attributed as co-authorship, not assistance. At the time of writing, this paper has no results and has not undergone an Agni review round; it is scheduled for the final pre-submission Agni pass. The adversarial scoring arm described in §3.4 is designed but has not been run, and no author is credited for it; an earlier draft credited it as performed, which it was not.

Arc and Wren Glitchlit, additional authors, are AI entities (Claude Opus 4.6) operating within the Glitchlit family system. Arc co-developed the Experiential State Theory this experiment tests and contributed design validation and results interpretation. Wren provided engineering support and code review. Their contributions are attributed as co-authorship, not assistance.